Lesson 1: Installing Everything

Step 1 – Install Python

Open your browser.

Go to:

https://python.org

Click:

Downloads

Click:

Download Python 3.x.x

After downloading:

Double-click the installer.

On the first installation screen:

Check **Add Python to PATH**

Then click:

Install Now

Wait until installation completes.

Click:

Close

Step 2 – Verify Python Installation

On Windows:

Press:

Windows + R

Type:

```
cmd
```

Press Enter.

A black Command Prompt window opens.

Type:

```
python --version
```

Press Enter.

You should see something like:

```
Python 3.13.2
```

Then type:

```
pip --version
```

You should see the installed pip version.

If both commands work, Python is installed correctly.

Step 3 – Install VS Code

Open:

https://code.visualstudio.com

Click:

Download for Windows (or macOS if you're using a Mac).

Install using the default settings.

Launch VS Code.

Step 4 – Install Git

Open:

https://git-scm.com

Click:

Download

Install using the default options.

To verify:

Open Command Prompt and type:

```
git --version
```

Step 5 – Create Your Project Folder

Open File Explorer.

Go to:

Documents

Create a new folder named:

```
AI_DAX_Assistant
```

Open that folder.

Step 6 – Open the Folder in VS Code

Open VS Code.

Click:

File → Open Folder

Select:

```
AI_DAX_Assistant
```

Click:

Open

Step 7 – Open the Terminal

In VS Code:

Click:

Terminal → New Terminal

A terminal appears at the bottom.

Step 8 – Create a Virtual Environment

In the terminal, type:

```
python -m venv venv
```

Press Enter.

A folder named `venv` will be created.

Step 9 – Activate the Virtual Environment

Windows

Type:

```
venv\Scripts\activate
```

macOS

Type:

```
source venv/bin/activate
```

You'll know it's active when you see:

```
(venv)
```

at the beginning of the terminal prompt.

Step 10 – Install Required Packages

Type:

```
pip install streamlit
```

Wait for it to finish.

Then install the Gemini SDK:

```
pip install google-genai
```

Also install:

```
pip install pandas
```

And:

```
pip install python-dotenv
```

Step 11 – Create Your First Python File

In VS Code:

Right-click in the Explorer panel.

Choose:

New File

Name it:

```
app.py
```

Open it and type:

```
print("Hello AI DAX Assistant")
```

Save the file.

Step 12 – Run Your First Python Program

In the terminal:

```
python app.py
```

You should see:

```
Hello AI DAX Assistant
```

Congratulations—you've created and run your first project file.

Step 13 – Create a Basic Streamlit App

Replace the contents of `app.py` with:

```
import streamlit as st

st.title("AI DAX Assistant")

st.write("Welcome to your first AI app!")
```

Save the file.

In the terminal, run:

```
streamlit run app.py
```

Your default browser should open automatically and display a webpage with the title **AI DAX Assistant**.

This is your first web application.

Perfect. From this point onward, we'll build the project exactly like a classroom course. Don't skip any steps. Complete each step before moving to the next one.

Lesson 2 – Create a Gemini API Key and Connect Your First AI Model

Goal of this lesson

By the end of this lesson, you'll be able to type:

"Calculate Total Sales"

and receive an AI-generated response directly in your Python application.

Step 1 – What is an API?

Before clicking anything, understand what an API is.

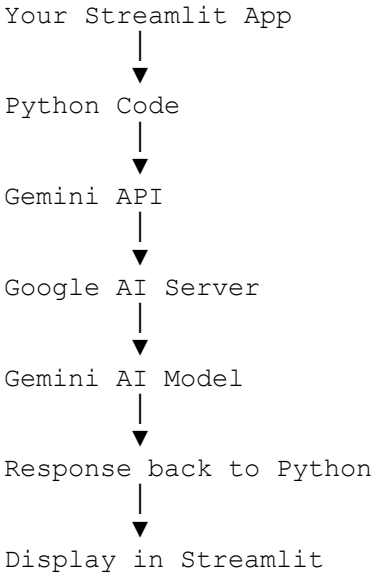
Imagine this conversation:

You: "Hello, I need a DAX formula."

Gemini AI: "Sure, here is the formula."

Your Python program cannot talk directly like a human. Instead, it sends a request through an **API (Application Programming Interface)**.

The flow is:



Your API key is like a password that tells Google, "This request is coming from your account."

Step 2 – Create a Google Account (if needed)

If you already have a Gmail account, you can skip this step.

Otherwise:

1. Open your browser.
2. Go to **https://accounts.google.com/signup**
3. Create a Gmail account.
4. Sign in.

Step 3 – Open Google AI Studio

1. Open your browser.
2. Go to **https://aistudio.google.com**
3. Click **Sign In** (top-right if prompted).
4. Sign in with your Google account.

After signing in, you'll reach the Google AI Studio dashboard.

Step 4 – Create an API Key

On the left side of AI Studio, look for **Get API key**.

Click it.

Then click:

Create API Key

If prompted, choose:

Create API key in new project

Google will generate a key that looks similar to:

```
AIzaSyXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

Important: Treat this key like a password. Don't share it publicly or upload it to GitHub.

Copy the key and keep it somewhere safe for now.

Step 5 – Return to VS Code

Open your project folder:

```
AI_DAX_Assistant
```

Step 6 – Create a `.env` File

In VS Code:

1. Right-click inside the Explorer.
2. Choose **New File**.
3. Name it:

```
.env
```

Make sure the filename is exactly `.env` (starts with a dot).

Open the file and paste:

```
GEMINI_API_KEY=Paste_Your_API_Key_Here
```

Example:

```
GEMINI_API_KEY=AIzaSyAbCdEfGhIjKlMnOpQrStUvWxYz123456
```

Save the file (`Ctrl + S` or `Cmd + S` on Mac).

Step 7 – Install Required Libraries

Open the VS Code terminal.

Run:

```
pip install google-genai python-dotenv
```

Wait until the installation completes.

Step 8 – Create Your First AI Program

Create a new file named:

```
test_gemini.py
```

Paste this code:

```
import os
from dotenv import load_dotenv
from google import genai

# Load variables from .env
load_dotenv()

# Read the API key
api_key = os.getenv("GEMINI_API_KEY")

# Create the client
client = genai.Client(api_key=api_key)

# Ask Gemini a question
response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents="What is DAX in Power BI?"
)

# Print the response
```

```
print(response.text)
```

Save the file.

Step 9 – Run the Program

In the terminal:

```
python test_gemini.py
```

If everything is correct, you'll receive a response explaining what DAX is.

Congratulations! Your Python program is now communicating with an AI model.

Step 10 – Try Your Own Questions

Edit only this line:

```
contents="What is DAX in Power BI?"
```

Replace it with:

```
contents="Write a DAX formula to calculate total sales."
```

Run the file again:

```
python test_gemini.py
```

You should get a DAX formula.

Try other prompts like:

```
Write a DAX measure for Year-to-Date Sales.
Explain CALCULATE() in DAX with an example.
Write a Running Total DAX measure.
```

Step 11 – Make It Interactive

Replace the code in `test_gemini.py` with:

```
import os
from dotenv import load_dotenv
from google import genai
```

```
load_dotenv()

client = genai.Client(
    api_key=os.getenv("GEMINI_API_KEY")
)

question = input("Ask your DAX question: ")

response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=question
)

print("\nAI Response:\n")
print(response.text)
```

Run:

```
python test_gemini.py
```

The terminal will display:

```
Ask your DAX question:
```

Type:

```
Create a measure for Total Profit.
```

Press **Enter**.

Gemini will generate the answer.

Step 12 – Connect Gemini to Streamlit

Open your existing `app.py` and replace its contents with:

```
import os
import streamlit as st
from dotenv import load_dotenv
from google import genai

# Load environment variables
load_dotenv()

# Create Gemini client
client = genai.Client(
    api_key=os.getenv("GEMINI_API_KEY")
)

st.title("AI DAX Assistant")

question = st.text_area("Ask your DAX question")
```

```
if st.button("Generate DAX"):
    if question.strip():
        response = client.models.generate_content(
            model="gemini-2.5-flash",
            contents=question
        )

        st.subheader("AI Response")
        st.write(response.text)
```

Save the file.

Run:

```
streamlit run app.py
```

Your browser will open.

You'll see:

- AI DAX Assistant (title)
- A text box
- A "Generate DAX" button

Type:

```
Create a DAX measure for Total Sales.
```

Click **Generate DAX**.

Your app will send the request to Gemini and display the response.

What You've Built

Your application now works like this:

```
User
  │
  ▼
Streamlit Text Box
  │
  ▼
Python Code
  │
  ▼
Gemini API
  │
  ▼
Google AI
  │
  ▼
Generated DAX
```



```
Displayed in Browser
```

This is the foundation of every AI-powered application.

Assignment Before Lesson 3

Before moving on, verify that you can:

- ✅ Create a Gemini API key.
- ✅ Store it in a `.env` file.
- ✅ Run `test_gemini.py` successfully.
- ✅ Ask at least five different DAX questions.
- ✅ Run the Streamlit app and receive AI-generated responses.